PERIODIC PROPERTIES 2

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Periodic properties are those that change in an ordered manner, progressing through the Periodic Table. The size of atoms is an example of a periodic property. Note that the size of an atom can vary due to outside effects on the electron cloud. Figure 1 shows the average proportional sizes of atoms for part of the Periodic Table.

1 Name two Groups of elements that are missing from figure 1.

2 Describe what happens to the size of an atom, in general, moving:

- **a** left to right across a Period
- **b** down a Group.

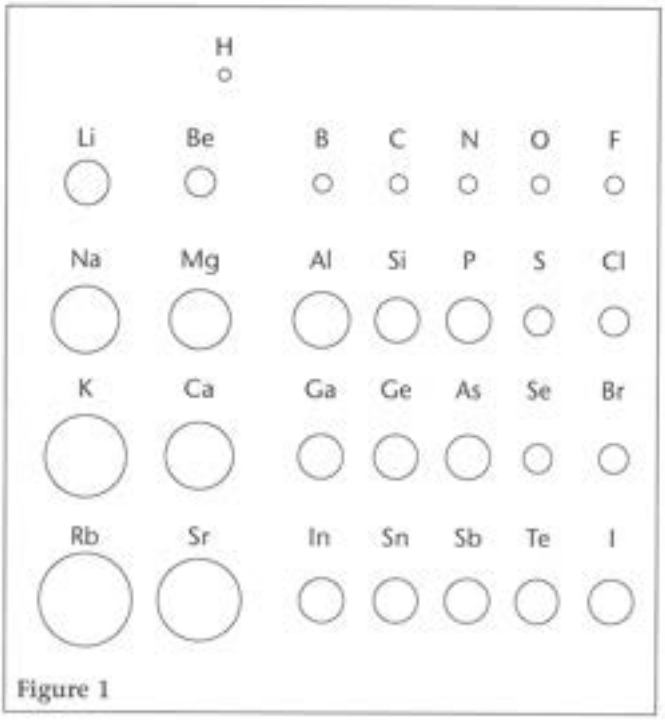


Figure 1

As you move across a Period from left to right, extra protons are being added to the nucleus and extra electrons to the outer electron shell. The increased attraction between the positive nucleus and electrons causes the electrons to be pulled closer to the nucleus.

3 Explain why atomic sizes increase moving down a Group.

4 Use the values in table 1 to draw a graph of atomic number (x-axis) against melting point (y-axis). Note that temperatures are in Kelvins (K). To change to degrees Celsius (°C), add 273 to the Kelvin figure. The melting point quoted for carbon is its sublimation temperature.

Element	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al
Melting point (K)	204	4	454	1551	2573	3573	77	55	53	27	371	924	933
Atomic no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Element	Si	P	S	Cl	Ar	K	Ca	Ga	Ge	As	Se	Br	Kr
Melting point (K)	1683	317	390	239	155	337	1115	303	960	304	490	266	116
Atomic no.	14	15	16	17	18	19	20	31	32	33	34	35	36

Table 1

Although there are some discontinuities (interruptions to the pattern), some trends should be visible.

5 Describe what happens to melting points moving down the elements in:

- **a** Group II
- **b** Group IV
- **c** the noble gases.

The type of bonding present in the elements influences the structure and hence the melting points of elements. Major structures include metallic, covalent lattice (macromolecular) and discrete molecular (single, separate molecules). Metallic bonding is usually associated with high melting points.

6 Apart from a high melting point, what other physical property could be associated with a typical metal?

7 Carbon forms macromolecules (e.g. graphite and diamond). Name one other element that is likely to form a similar structure. Give reasons for your answer.

8 Which Group of elements exist as discrete atoms?

9 Explain why atomic sizes decrease from left to right across a Period.

10 Name the element that has the lowest melting point.

11 Name the element that has the highest melting point.

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